

RECIPROCAL AI-ENABLED COMMUNITY-ENGAGED LEARNING IN ENVIRONMENTAL DATA ANALYTICS

A. Marsellos¹, K. Tsakiri²

¹Hofstra University, Geology, Environment, and Sustainability (New York, UNITED STATES)

²Rider University, Information Systems, Analytics, and Supply Chain Management (New Jersey, UNITED STATES)

Abstract

Artificial intelligence tools are increasingly integrated into higher education; however, there remains limited empirical research examining how AI-assisted analytical methods function within sustained, community-engaged experiential learning environments. This presentation introduces the Community-Engaged Earth Data and Sustainability (CEEDS) model, an AI-enabled instructional framework implemented in an undergraduate geoscience course in collaboration with local high school partners. The CEEDS model builds upon prior experience from the NSF-funded GEOTeams program (IUSE #1911514), which employed multilevel research teams to broaden participation in geoscience education. Extending this foundation, CEEDS integrates RStudio programming, authentic environmental datasets (e.g., earthquake, groundwater, and climate data), and structured, reciprocal research collaboration between university students and high school students. Undergraduate participants receive training in AI-assisted data analysis and scientific communication before engaging in small mixed research teams that co-develop sustainability-focused projects and produce interactive dashboards and public presentations. Using a mixed-methods evaluation design, the project measures growth in AI literacy, programming competency, collaborative research confidence, and mentorship effectiveness through anonymous pre- and post-course surveys, rubric-based assessments, and coded qualitative reflections. Preliminary findings suggest that embedding AI tools within reciprocal research partnerships enhances student engagement and promotes deeper understanding of data-driven environmental analysis.

Keywords: Artificial Intelligence in Education, Experiential Learning, Community-Engaged Learning, STEM Education, Technology-Enhanced Pedagogy.

1 INTRODUCTION

Artificial intelligence (AI) is increasingly integrated into higher education, yet its pedagogical impact depends fundamentally on how it is embedded within learning environments. While AI is often framed as a tool for efficiency or automation, its deeper educational value emerges when it is situated within authentic, collaborative, and inquiry-driven contexts [1], [2]. The rapid advancement of artificial intelligence (AI) has necessitated its integration into the modern higher education curriculum. While many institutions have begun "Developing AI Literacy" as a priority, there remains a significant gap in the literature regarding how these tools influence collaborative research skill development within reciprocal, community-engaged contexts.

Recent studies further highlight regional disparities in the adoption of data-driven geoscience education across the United States. For example, Phan and Marsellos [3] found that institutions in Western states are more actively integrating advanced Earth data analytics and AI-based methodologies into their curricula, whereas institutions in Eastern states tend to adopt more gradual and conservative approaches, with fewer programs offering courses focused on AI-enabled geoscience analysis. These findings underscore the need for scalable and adaptable instructional models, such as CEEDS, that can bridge this gap and expand access to advanced analytical training.

At the same time, research in STEM education consistently identifies experiential learning, undergraduate research, and community-engaged pedagogy as high-impact practices that enhance student learning, persistence, and identity formation [4], [5]. These approaches are particularly important in geoscience education, where workforce shortages and limited diversity continue to present significant challenges.

The GEOTeams model addressed these challenges by introducing a multilevel team-based research framework involving high school students, undergraduate students, and educators working collaboratively on authentic geoscience problems [6]. This model demonstrated that structured

mentorship and teamwork can foster engagement, broaden participation, and support meaningful entry into geoscience pathways.

This study introduces and evaluates the Community-Engaged Earth Data and Sustainability (CEEDS) model. The project builds upon interdisciplinary foundations, including the Novel Interdisciplinary Teaching Model for AI-Driven Research in Geoscience and Business presented at EDULEARN 2025 [7]. The CEEDS framework formalizes a high-impact approach designed to move beyond theoretical knowledge toward measurable, professional research output. It also reflects the continuing interdisciplinary collaboration between Hofstra University and Rider University in AI-supported teaching and research. By embedding AI-assisted analysis within structured, reciprocal research teams, students are required to interpret real-world environmental datasets and disseminate findings with societal relevance.

Importantly, prior work has shown that when students are embedded in authentic research environments with strong mentorship, they are capable of producing high-quality scholarly outputs. For example, undergraduate-led work has resulted in peer-reviewed publications employing advanced analytical techniques such as time series regression for flood forecasting [8].

Further evidence comes from international multilevel research collaborations, where students participated in volcanic monitoring projects involving field instrumentation, drone thermal imaging, and environmental data analysis [9]–[11]. These experiences included NSF-supported high school students working alongside undergraduate and graduate researchers, faculty mentors, and international collaborators. These pathways have demonstrated sustained impact, including student-led conference presentations and progression into advanced geoscience programs [12], [13].

The purpose of this study is to systematically investigate the pedagogical impact of a reciprocal, AI-enabled, community-engaged experiential learning model on student learning outcomes, AI literacy, collaborative competence, and engagement. Building on the CEEDS framework, the study is guided by the following research questions:

- (1) How does participation in a reciprocal, AI-enabled community-engaged learning model influence university-level students' RStudio programming competency, AI literacy, and collaborative research confidence?
- (2) How does participation affect high school students' confidence in data analysis, perceptions of access to advanced STEM learning, and engagement in sustainability research?
- (3) What qualitative themes emerge regarding mentorship effectiveness, reciprocity, and community impact within a sustained, cross-institutional experiential learning framework?

These questions align with the broader objective of evaluating how AI-assisted analytical tools can be meaningfully embedded within authentic, collaborative learning environments that extend across institutional and educational boundaries.

2 METHODOLOGY

The CEEDS model utilizes a systematic, pedagogical framework adapted from successful pilots in interdisciplinary AI education. The methodology focuses on four core pillars: experiential learning, high-impact practices, reciprocity, and community-engaged STEM education.

2.1 Experiential learning and authentic research

Experiential learning emphasizes active engagement with real-world problems. In STEM education, this often takes the form of authentic research, where students analyze data, test hypotheses, and interpret results. Such environments promote deeper understanding and sustained engagement [4]. Crucially, authentic research experiences have been shown to produce publishable-quality outcomes when supported by structured mentorship, as demonstrated in undergraduate contributions to peer-reviewed geoscience studies [7].

2.2 High-impact educational practices

High-impact practices such as collaborative research, iterative feedback, and public dissemination, are associated with strong learning gains and student persistence [2]. CEEDS incorporates these elements by requiring students to work in teams, develop analytical outputs, and present findings

publicly.

2.3 Reciprocity and multilevel mentorship

The CEEDS model emphasizes reciprocal learning across educational levels. This structure builds directly on GEOTeams [6], where collaboration among high school students, undergraduates, and educators supported both technical learning and community building. Evidence from subsequent implementations shows that such models can extend into co-authored research and sustained academic engagement [8]–[12].

The implementation involves undergraduate and graduate students collaborating with high school students. In this reciprocal model, university students serve as mentors while co-developing projects with their high school counterparts. The framework relies on RStudio or Python as the core platform for data analysis, visualization, and the application of machine learning methods such as Random Forest and clustering.

2.4 Community-engaged sustainability education

Environmental data analytics inherently connects scientific inquiry with societal relevance. CEEDS situates data analysis within sustainability contexts, allowing students to explore real-world issues such as groundwater variability, climate patterns, and natural hazards at local and international level.

2.5 CEEDS as a pedagogical model

The CEEDS model translates the theoretical framework into a structured learning process that integrates AI-assisted analysis with experiential and community-engaged learning. The model is designed as a progression from guided instruction to independent, collaborative research, where students actively engage with real-world data and develop analytical and communication skills in a supportive team environment.

The implementation begins with the introduction of authentic environmental datasets, including earthquake records, groundwater measurements, and climate-related time series. These datasets provide a foundation for inquiry and allow students to explore sustainability-related questions at both local and global scales. Students are then introduced to computational tools such as RStudio or Python, where they develop programming skills alongside the use of AI-assisted methods for data analysis, visualization, and model exploration. Coding serves as an accessible entry point for working with real data, enabling students to quickly perform analysis and present results in the short term. This progression is particularly important given that geoinformatics and data-driven coursework remain limited in a study of high-ranked U.S. institutions, especially in more traditional academic settings [3], reinforcing the need for models that provide early exposure to data analytics and AI-supported methods. Over time, this experience fosters curiosity and motivates students to seek and collect their own data through field-based research.

Following this initial training, students transition into mixed research teams composed of participants from different educational levels. Within these teams, students develop research questions, explore multiple analytical approaches, and interpret results using AI-supported workflows. This stage reflects the core of the CEEDS model, as it combines technical skill development with mentorship, peer learning, and iterative problem-solving. The presence of high school students reinforces the reciprocal nature of the model, requiring undergraduate participants to communicate complex ideas clearly while supporting less experienced collaborators.

A distinguishing feature of CEEDS is its extension beyond the classroom into applied research environments. In particular, selected implementations include participation in international field-based projects, such as volcanic monitoring in the Aegean region. In these settings, students engage in data collection, instrumentation monitoring, and real-time environmental analysis in collaboration with international research partners. These experiences connect computational analysis with physical data acquisition and reinforce the relevance of environmental data science.

As illustrated in Fig. 1 and Fig. 2, these field-based activities bring together undergraduate students, graduate students, faculty mentors, high school participants, and community members within a shared research framework. This structure reflects an expansion of the GEOTeams model into a global and

applied context, where students participate in authentic scientific collaboration across educational levels and institutional boundaries.

The final stage of the CEEDS model focuses on dissemination, where students present their findings through dashboards, presentations, and research outputs. This component reinforces learning by requiring students to synthesize results, communicate effectively, and connect their work to broader societal and environmental challenges.

2.6 Research Design

A mixed-methods research design is employed to evaluate the pedagogical impact of the model. Quantitative data collection includes anonymous pre- and post-course surveys measuring: RStudio programming competency, AI literacy and responsible use of generative AI tools, and collaborative research self-efficacy and confidence between High School students, university students, and faculty. Qualitative data includes structured student reflection essays and partner feedback, which are coded to identify recurring themes regarding mentorship effectiveness and the perceived societal relevance of the research.

3 RESULTS

Preliminary results from the pilot implementation of related AI-driven pedagogy (e.g., the BUS444 course) indicate that structured, conference-driven models significantly enhance student productivity and professional scholarly communication.

3.1 Preliminary outcomes

Preliminary implementation of the CEEDS model and its predecessor frameworks has produced a range of measurable academic and experiential outcomes that reflect the effectiveness of reciprocal, community-engaged learning structures.

Students participating in multilevel research teams have contributed to conference presentations at institutional, regional, and international levels, including work on environmental monitoring, climate variability, and geospatial analysis [8]–[12]. These outputs demonstrate the ability of students, including high school participants, to engage in meaningful scientific communication and dissemination.

In addition, collaborative research conducted in international field settings—particularly in active volcanic systems in the Aegean region—has provided students with hands-on experience in data collection, instrumentation, and thermal drone analysis [8]–[10]. These field experiences have extended classroom learning into real-world scientific environments and have supported the development of technical and analytical competencies.

The model has also facilitated co-authorship and continued research engagement, with students contributing to ongoing projects beyond their initial participation. In several cases, high school students have continued presenting research at regional conferences and pursuing academic pathways in geoscience and related fields [11], [12]. These trajectories suggest that early exposure to authentic research environments, combined with mentorship and collaborative learning, can significantly influence long-term academic and professional development.

Overall, these outcomes provide preliminary evidence that the CEEDS model supports not only skill development in AI-assisted data analysis but also broader educational goals, including student engagement, confidence, and access to advanced STEM pathways.

3.2 AI-Enabled Productivity Across Disciplines

Evidence from prior iterations shows that students who engage in AI-assisted methodologies produce higher-quality research outputs suitable for external academic venues. For instance, students utilizing this approach successfully produced multiple conference publications focused on applied AI-driven analytics, such as forecasting industry growth and evaluating ethical limits in decision-making. These efforts demonstrate how students elevated their presentation skills from high school to academic performance and quality with strong commitment, especially to innovative ideas and applications. In particular, one team of students from Rider University (course AI in Business) created their own mobile

app [15]. The next project focused on global emissions and their ethical limits for sustainable decision-making. This research utilized K-means clustering and random forest models in RStudio to analyze global datasets on greenhouse gas emissions, trade, and transportation, grouping 148 countries into low, intermediate, and high profiles based on GDP, emissions, and mobility [16]. A key finding was that while AI models successfully identified patterns, their outputs were fundamentally shaped by the human-defined ethical frameworks used to label sustainability, underscoring the critical role of human oversight.

The research by Hofstra students (course Geohazards) Izaguirre Cortez, Adalia Rezki, and Najarro Lima further explores these ethical dimensions by applying AI and machine learning to sustainable infrastructure through a Hurricane Sandy modeling perspective [14]. This study investigates how automated decision-making processes can be aligned with ethical frameworks to improve the resilience of coastal infrastructure against extreme weather events. By analyzing historical data from Hurricane Sandy, the project demonstrates that while machine learning can significantly enhance predictive accuracy for storm impacts, the resulting decision-making models must be grounded in human-defined ethical considerations to ensure equitable and sustainable outcomes for affected communities.

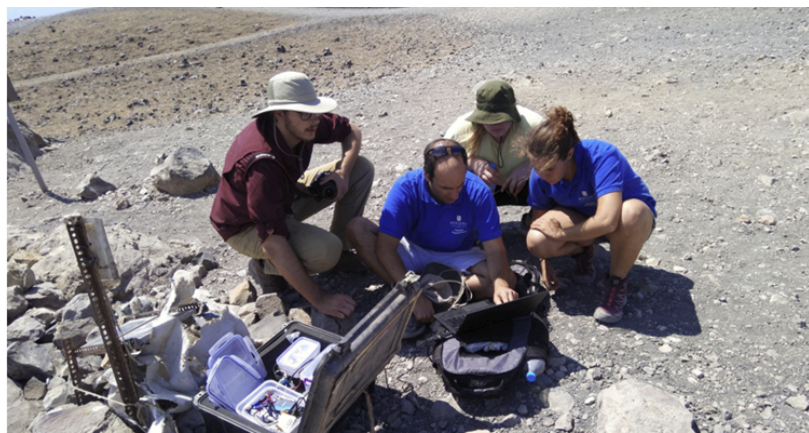


Figure 1. Multilevel research team conducting field-based data analysis during environmental monitoring activities in a volcanic setting (Santorini Volcano, Greece). The team includes undergraduate students, a graduate student, and faculty mentorship, demonstrating collaborative, technology-supported analysis in an authentic research environment.



Figure 2. Community-engaged field research participation in an international volcanic monitoring project (in Sousaki Volcano, Greece). The group includes undergraduate students, a high school student, faculty mentorship, and family support, illustrating the broader ecosystem of reciprocal engagement and access to advanced geoscience research experiences.

4 DISCUSSION

The findings and preliminary outcomes associated with the CEEDS model suggest that artificial intelligence can support learning most effectively when embedded within authentic, collaborative, and community-engaged environments. Rather than serving primarily as a tool for efficiency, AI in this context becomes part of a broader pedagogical framework that supports inquiry, interpretation, and communication.

A central contribution of the CEEDS model is its emphasis on reciprocal, multilevel collaboration. Building on the National Science Foundation (NSF) funded GEOTeams framework [6], structured interaction between high school students, undergraduate students, and faculty fosters a learning environment where knowledge is co-constructed. Undergraduate students deepen their understanding through mentorship, while high school students gain access to advanced analytical tools and research practices. This reciprocal structure strengthens both technical competence and academic confidence across participant groups.

The cross-institutional dimension, including collaboration between institutions in different geographic regions, further enhances student learning. Exposure to multiple academic environments broadens students' understanding of research cultures and STEM pathways, particularly for high school participants. This aligns with the study's focus on access, engagement, and perceptions of higher education opportunities.

The CEEDS model also highlights the importance of authentic research experiences in supporting high-level student outcomes. Participants have contributed to conference presentations and, in some cases, peer-reviewed research [8]–[13], reinforcing findings from prior studies on high-impact educational practices [4], [5]. These outcomes demonstrate that structured mentorship combined with real-world data analysis can enable students to engage in meaningful scientific work.

An important pedagogical insight relates to communication challenges in AI-assisted learning. Many students show difficulty in formulating clear and structured queries when interacting with AI tools, reflecting broader shifts in learning behaviors following remote instruction during the COVID-19 period. Since effective AI use depends on precise input, these communication limitations can affect analytical outcome.

Despite its strengths, the CEEDS model also presents challenges that must be acknowledged. One of the most common issues observed is the lack of clear communication and transparency within team-based environments, particularly when participants come from different educational levels and backgrounds. As in real-world collaborative settings, teamwork requires time, mutual understanding, and the ability to navigate differences in expectations, experience, and communication styles. In some cases, students struggled to articulate ideas clearly or to align their contributions with the overall research direction. However, these challenges also represent an essential component of the learning process. The CEEDS model intentionally provides a degree of flexibility that allows students to explore ideas that may not always directly align with the initial research objectives. By fostering an environment where diverse perspectives are respected and encouraged, students gradually develop the ability to take initiative, assume responsibility, and contribute innovative approaches. This balance between structure and intellectual freedom is critical for supporting both effective collaboration and creative problem-solving.

The CEEDS model addresses this challenge by embedding AI use within team-based collaboration, where students continuously refine how they ask questions, interpret outputs, and communicate findings. Interaction with peers, mentors, and community participants helps students develop both technical and communication skills. In this context, communication becomes a critical component of both human collaboration and AI interaction. In several instances, students also engaged with AI tools to seek guidance on how to approach audience engagement, present their ideas, or navigate collaboration within their teams. While such interactions require careful interpretation, they often encourage students to pause, reflect, and consider alternative perspectives before acting. This process contributed to more thoughtful decision-making and reduced impulsive responses, ultimately supporting a more stable and constructive collaborative environment.

The inclusion of international field-based research further strengthens the model by extending learning beyond the classroom. Participation in volcanic monitoring projects provides students with hands-on experience in data collection and real-time analysis, reinforcing the connection between analytical work and real-world applications. The involvement of high school students in these settings demonstrates the potential of such models to expand access to advanced scientific experiences.

At the same time, the use of AI-assisted tools requires careful instructional guidance. Without proper support, students may rely too heavily on automated outputs or lack sufficient understanding of analytical methods. CEEDS addresses this by emphasizing critical evaluation, requiring students to interpret and validate results within a scientific framework.

Overall, the CEEDS model demonstrates that the effectiveness of AI in education depends on its integration within collaborative, research-oriented environments. By combining technical skill development with mentorship, communication, and real-world relevance, the model offers a framework for designing inclusive and impactful AI-enabled learning experiences.

5 CONCLUSIONS

The CEEDS model shows that AI can be most effective in education when it is used within collaborative, real-world learning environments. By combining teamwork, mentorship, and hands-on data analysis, the model helps both university and high school students build technical skills, improve their understanding of AI, and gain confidence in research. The findings address the study's research questions by demonstrating improvements in programming competency, AI literacy, collaborative confidence, and student engagement across educational levels. At the same time, the results highlight the importance of communication. Students need to learn how to ask better questions, evaluate different AI-generated results, and make thoughtful decisions. Working in teams and engaging with real research problems helps strengthen these skills. The inclusion of international fieldwork further connects classroom learning to real scientific applications, making the experience more meaningful for students. While challenges related to communication and teamwork remain, they are also part of the learning process and contribute to the development of critical collaboration skills. Overall, the CEEDS model provides a practical and adaptable approach for integrating AI into STEM education through collaboration, mentorship, and real-world engagement.

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